

It is important to give justifications for all your answers.

1 Map-Reduce: Social Network

We consider a social network of users. Each user is identified by a number UID. We assume each user can send messages to other users and “like” the posts of other users. Logs of these events are stored in a set EVENT of tuples (EID, UID1, UID2, TYPE, DATE) where EID is the event ID (a number), UID1 and UID2 are identifiers of two users, TYPE can be either "like" if user1 liked a post of user2 or "msg" if user1 sent a message to user2 and DATE is the date of the message or like. For instance, the event of EID 1 in the table below is a message sent by user 1 to user 2.

We perform a map/reduce operation on this data using the following code:

```
Output = EVENT.map(m1).reduceByKey(r1)
```

and with the following functions:

```
m1(x)
  eid, uid1, uid2, type, date = x
  return((uid2, type), 1)
```

```
r1(v1, v2)
  return v1+v2
```

We assume that the EVENT set is stored without duplication on two nodes N1 and N2 as follow:

	On N1:					And on N2:				
	EID	UID1	UID2	TYPE	DATE	EID	UID1	UID2	TYPE	DATE
	1	1	2	"msg"	2021/07/12	7	2	1	"like"	2021/07/12
	2	1	3	"like"	2021/08/11	8	2	1	"like"	2021/08/05
	3	1	3	"msg"	2021/07/14	9	3	3	"msg"	2021/05/15
	4	2	3	"msg"	2021/09/21	10	3	2	"msg"	2021/10/22
	5	2	3	"like"	2021/10/03	11	1	2	"msg"	2021/09/03
	6	2	3	"msg"	2021/05/14	12	1	2	"msg"	2021/11/05

In the following questions, you will explain how this map/reduce is done on this particular dataset first assuming there is no “pre-reduce” optimization on the nodes N1 and N2 and then assuming it is the case.

First, We assume that no pre-reduce operations are done on N1 and N2. This means all the reduce calls are computed on another node N3.

Question 1 : (1 pt) How many map calls are done on each node N1 and N2?

Question 2 : (1 pt) Give the full list of pairs that are send on the network after the map by each node N1 and N2.

Question 3 : (1 pt) Then give all the reduce function calls that are done on N3 and the results of each of these calls (represent these calls in a tree like fashion).

Now, we assume that the optimization using “pre-reduce” calls is done on N1 and N2.

Question 4 : (1 pt) Explain what is computed by each node N1 and. Give the full list of pairs that are send on the network from each node N1 and N2. Draw all the calls to reduce operation in a tree like fashion.

next exercise on back

2 Map Reduce: Telecommunication

A Telecommunication Company called ATC has data about the calls made by its customers. This data consist of a set CALL of tuples (CID, IDC, Num, IDR, Dur, Date) where CID is the ID of the call, IDC is the ID of the customer making the call, Num is the phone number called, IDR is the customer ID of the recipient (or -1 if the recipient is not a customer of ATC), Dur is the duration of the call in seconds and Date is when the call was made. If IDR is -1 it means that the recipient of the call is not a customer of ATC.

For each question, you have to answer using map/reduce operations. You can use several map and/or reduce operations in any order. The only map/reduce operation allowed are `map`, `flatMap`, `reduceByKey` and `filter`.

Question 5 : (1 pt) Compute a RDD of tuples (CID, IDC, Num, IDR, Dur, Date) with only the calls internal to the ATC company (i.e., where the recipient is a customer of ATC).

Question 6 : (1 pt) Compute the total price of calls for each customer. The output must be a RDD of pairs (IDC, tot_price). Each call of duration d made by a customer has a cost $\text{price}(d)$. You can use the function `price()` in your functions.

Question 7 : (2 pts) Compute a RDD (IDC, IDR, Maxdur, Date) of tuples giving the longest call for each customer. Maxdur is the duration (in seconds) of the longest call made by customer IDC. IDR is the recipient of this longest call and Date is the date of this call.

3 Recommendation Systems

The following table represents a part of a utility matrix R giving the ratings of users (in rows) on different items (columns): r_{ij} is the rating of user i for item j .

If an entry of the matrix is empty, it means the user did not evaluate this item.

R	I1	I2	I3	I4	I5	I6	I7
User 1		2	1		5		4
User 2	3			2		3	1
User 3	4	1				3	5
...							

The following table gives the similarity (between 0 and 1) between items (the matrix is symmetric, so only the upper triangle is given).

Sim	I1	I2	I3	I4	I5	I6	I7
I1	1	0.5	0.3	0.7	0.4	0.5	0.8
I2		1	0.4	0.2	0.5	0.2	0.5
I3			1	0.6	0.6	0.4	0.2
I4				1	0.7	0.2	0.6
I5					1	0.4	0.1
I6						1	0.2
I7							1

We want to compute estimation with basic item-item collaborative filtering using 2 neighbors for the estimation.

Question 8 : (1 pt) Given these similarities and the ratings, compute an estimate of the rating r_{11} of user 1 for item 1. (explain)

Question 9 : (1 pt) We now want an estimate of the rating r_{23} of user 2 for item 3. Explain which items will be used in the estimation and why. (You do not need to compute the estimate.)